# Midiverse Modular

## MVM004U – Button Build Guide

Thanks for choosing this kit and supporting Midiverse Modular!

This guide provides basic instructions to build your MVM004U Button module.

Module size: 12HP
Power consumption: draws 28mA from the +12 rail

In addition to basic soldering equipment, you will need wire strippers.

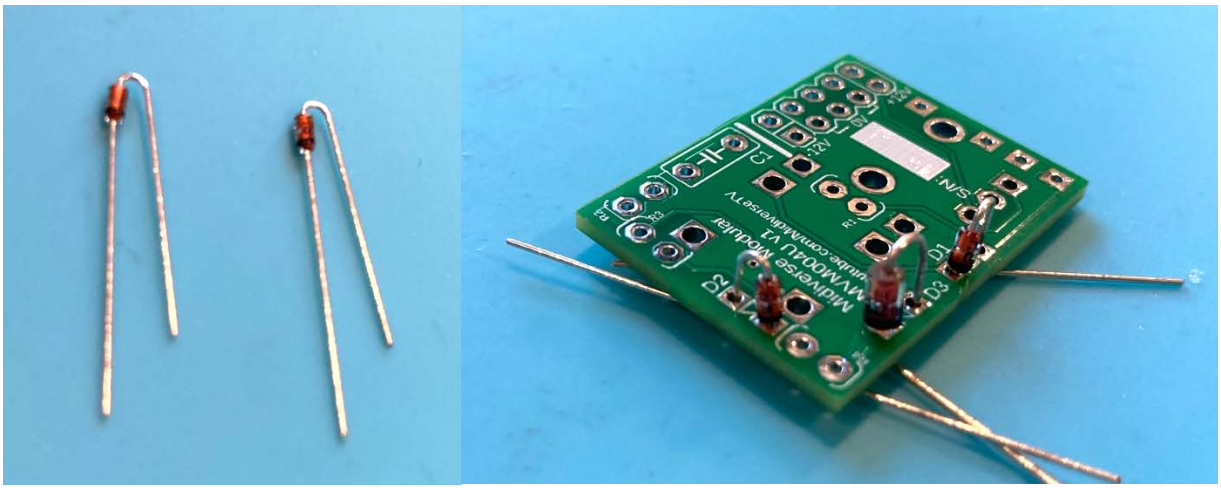



This kit comes with the following parts:

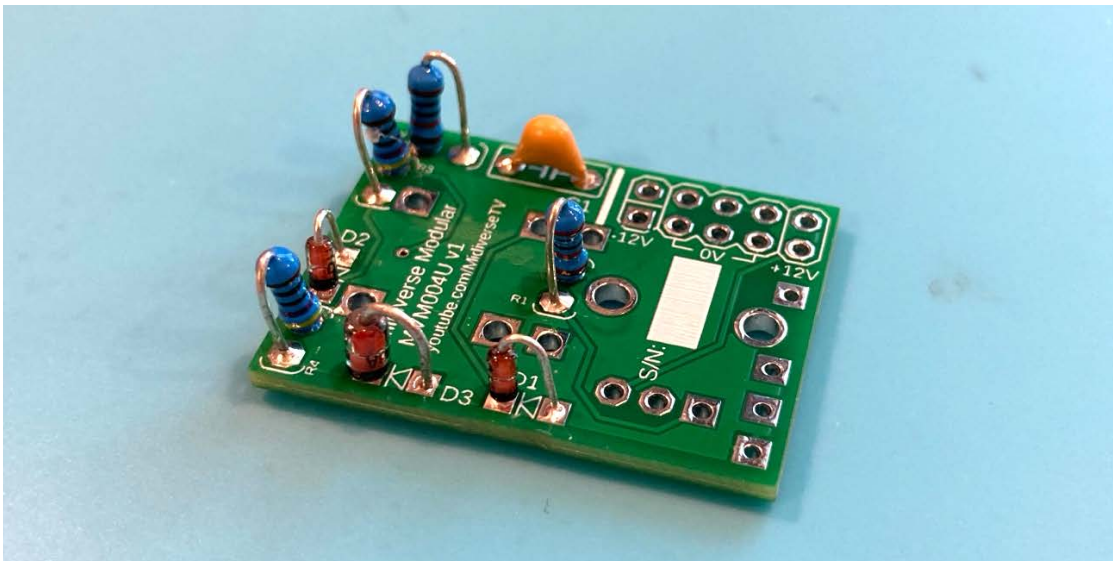
| Reference | Qty | Value | Notes |
|---|---|---|---|
| 1U Button PCB | 1 | | |
| 1U Button Panel | 1 | | |
| R3, R4 | 2 | 470R | Yellow, purple, black, black, brown |
| R2 | 1 | 10K | Brown, black, black, red, brown |
| R1 | 1 | 20K | Red, black, black, red, brown |
| D1, D2 | 2 | 1N4148 | |
| D3 | 1 | 1N4733 | |
| C1 | 1 | 0.1uf | Ceramic |
| IN, OUT | 2 | 3.5mm Jacks | |
| VR1 | 1 | B100K Pot | |
| J1 | 1 | 10 pin power connector | 2x5 header |
| Arcade button | 1 | | Momentary (blue) or latching (red) |
| Black wire | 2 | | |
| Red wire | 2 | | |
| White Knob | 1 | | |

**Build Instructions:**

Start by populating and soldering the 1N4148 and 1N4733 diodes in a vertical position. To do this, bend one leg to form a loop as in the photo below. Be sure to pay attention to the orientation. The black line on the diode should align with the white line on the PCB.



Next, populate and solder the resistors and the capacitor. The resistors should be populated in a vertical position similar to the diodes.

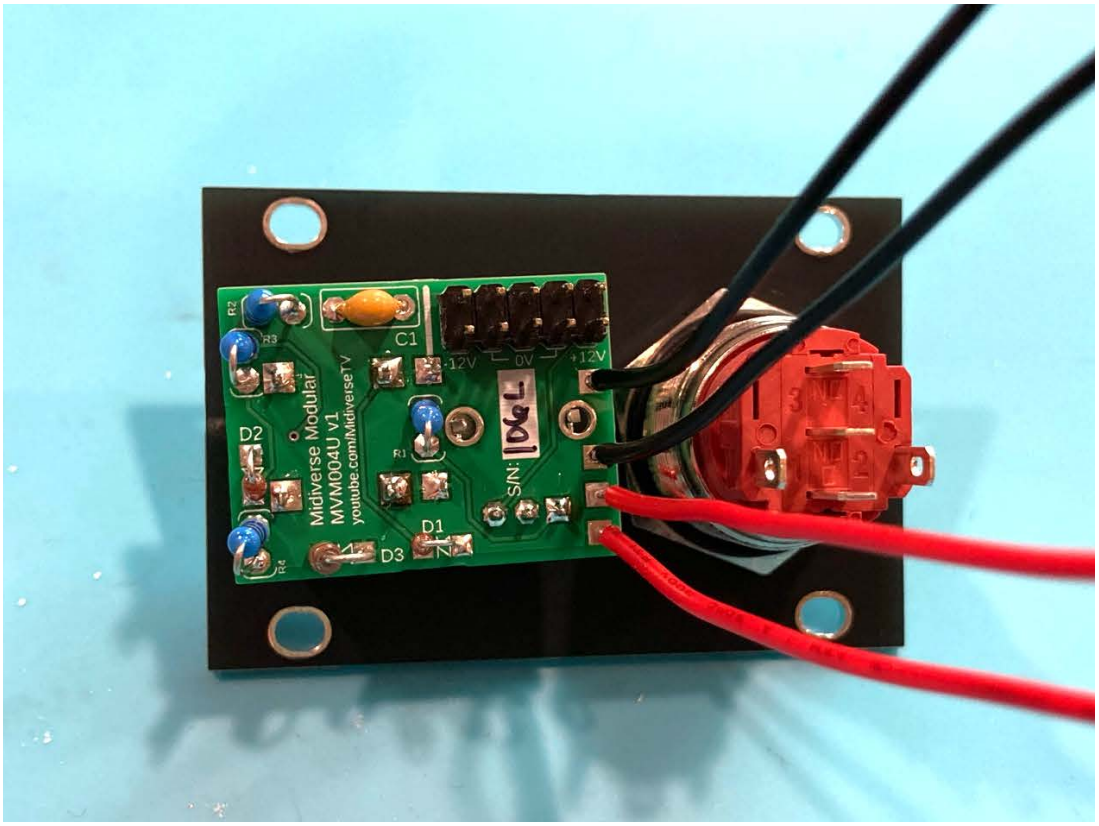



Next, solder in the 2 x 5 power connector. At this point, reflow all the solder joints.

Then, use the wire strippers to expose a small section of the wire and solder the wires to the individual pads as shown in the photo below. I group the black and red wires to orient myself when connecting them to the arcade button.

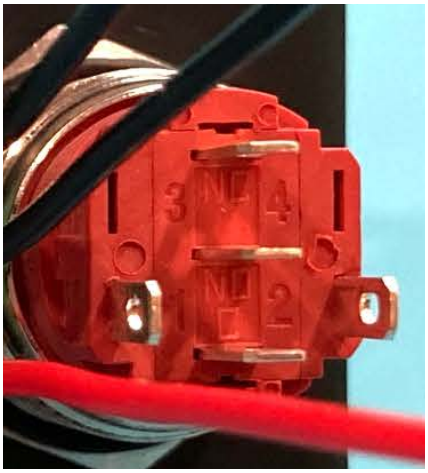
Populate the jacks and potentiometer, but **DO NOT** solder yet. **Before soldering the components**, be sure to attach the front panel by hand tightening the nuts. This step is very important as it is difficult to align these components with the holes if the front panel is not

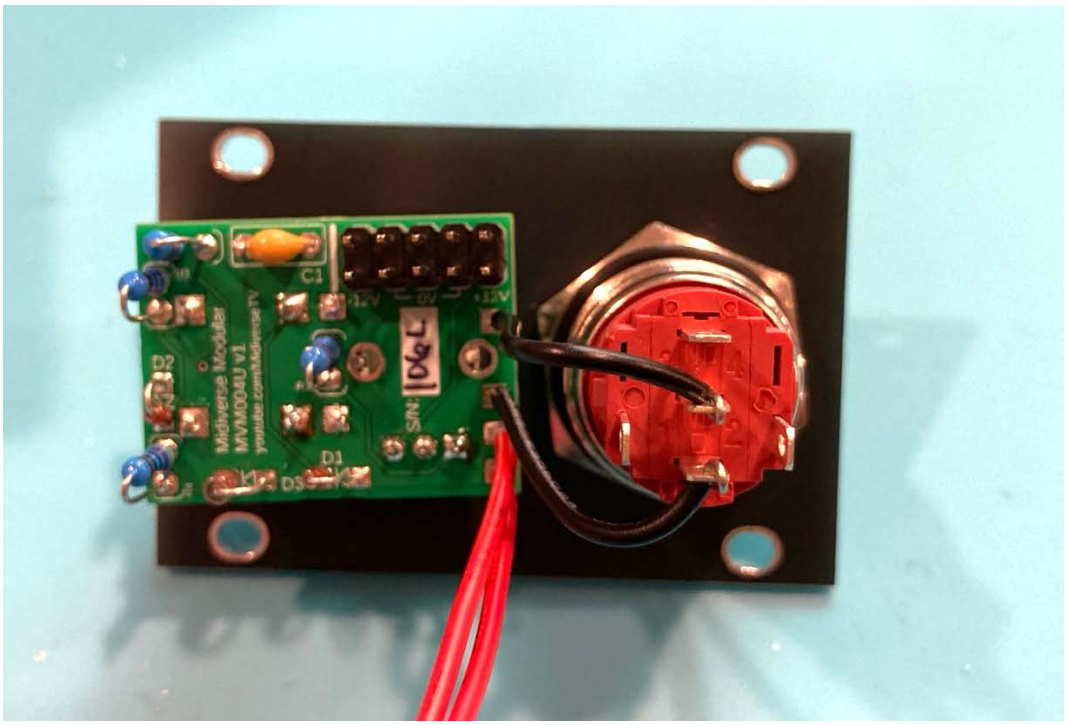
attached. Once the panel is attached, flip the module over and solder the jacks and potentiometer in place. Tighten down the nuts to secure the front panel. Reflow the solder joints.



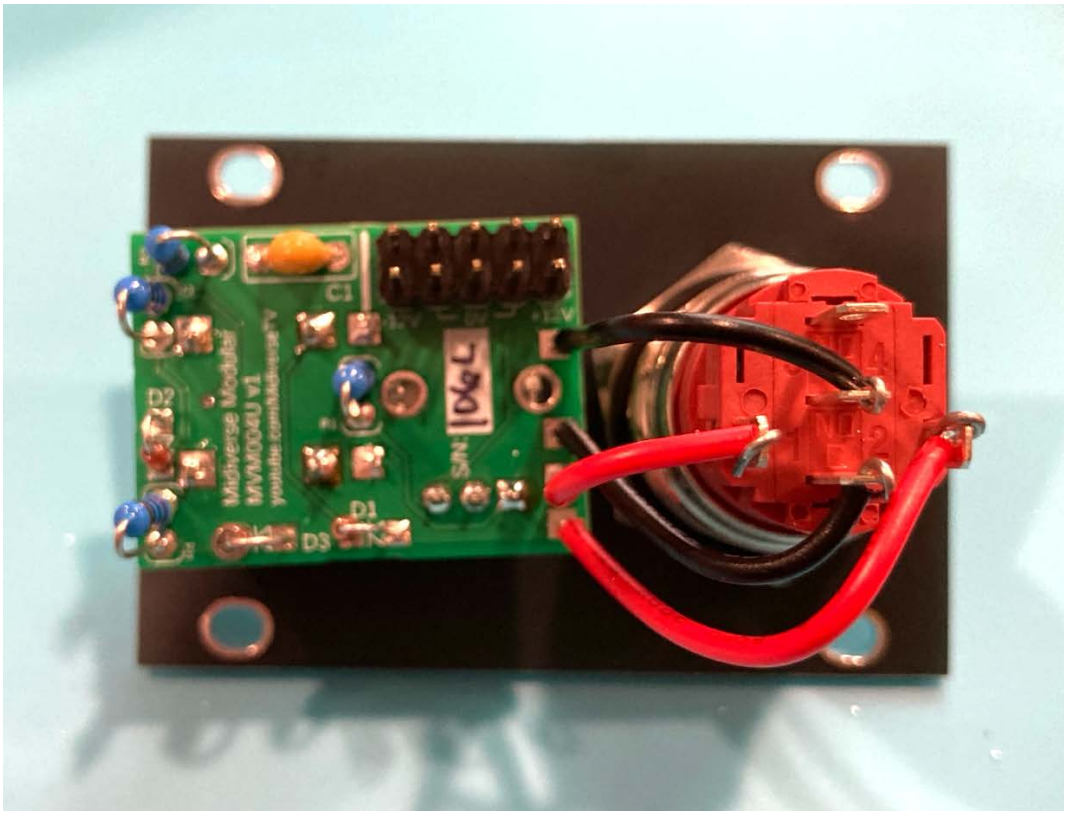

Connect the arcade button to the panel. Tighten the screw and then add the rubber washer (black ring) to secure it in place.

On the back of the button, the three pins in the middle are labeled NC, NO, and C. Connect the top two wires (just below the power connector; black wires in the photo below) to the **NO** and **C** pins.





Next, connect the bottom two wires (red wires in the photo below) to the two outer pins on the button, labeled 1 and 2.



Once the wires are secured to the pins, solder the wires in place. Be sure to not overheat the button.

Finally, put the white knob onto the pot. You’re finished!

